

PROJECT ASSIGNMENT 2

# Group10-PA2 Use Case Document

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FIFA.com and Chess.com

|               |                                                                                                              |
|---------------|--------------------------------------------------------------------------------------------------------------|
| <b>Course</b> | CSC13112 - UI/UX Design, FIT-HCMUS                                                                           |
| <b>Group</b>  | Group10                                                                                                      |
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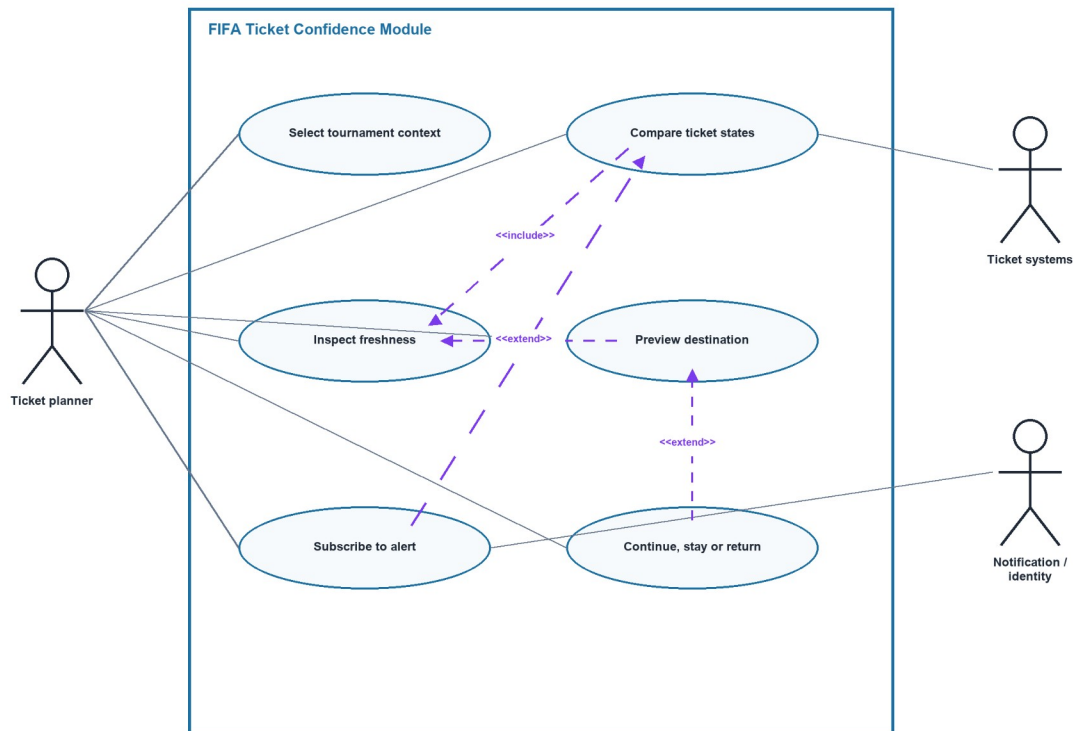
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# 1. Scope, actors, and notation

| System                                                                                                                                                                                                                                                                         | Primary actor                | External systems                                                                                                    | Use cases     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------|
| FIFA Ticket Confidence Module                                                                                                                                                                                                                                                  | Ticket planner               | FIFA tournament catalog; governed ticket-state feed; official ticketing partner; notification and identity services | F-UC01-F-UC06 |
| Chess Guided Beginner Review Module                                                                                                                                                                                                                                            | Beginner / returning learner | Game store; analysis service; explanation service; learning catalog                                                 | C-UC01-C-UC06 |
| <b>Notation:</b> Solid lines are actor associations. Dashed arrows labeled <<include>> point to required reused behavior. Dashed arrows labeled <<extend>> point from optional/conditional behavior to its base use case. External systems remain outside the system boundary. |                              |                                                                                                                     |               |

UC-UML-F FIFA Ticket Confidence Module

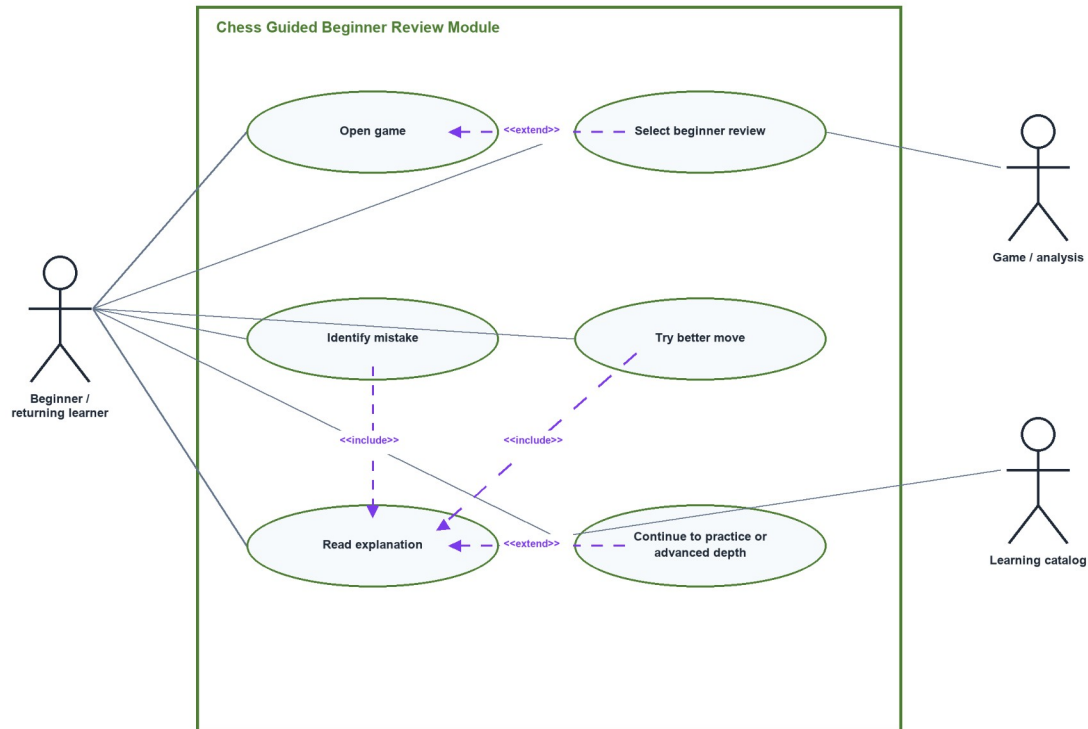


**Figure UC-UML-01.** FIFA UML use-case diagram

*Source: Design inference; actors and systems derived from F-A1 and the traceability matrix*

*Related: F-UC01-F-UC06*

## UC-UML-C Chess Guided Beginner Review Module

**Figure UC-UML-02.** Chess UML use-case diagram

Source: Design inference; actors and systems derived from C-A1 and the traceability matrix

Related: C-UC01-C-UC06

## 2. Detailed use cases

### 2.1 F-UC01 - Select tournament context

| Field            | Record                                                              |
|------------------|---------------------------------------------------------------------|
| ID               | F-UC01                                                              |
| Goal             | Select tournament context                                           |
| Scope            | FIFA Ticket Confidence Module                                       |
| Primary actor    | Ticket planner                                                      |
| Supporting actor | FIFA tournament catalog                                             |
| Trigger          | User opens the module from a tournament or ticket entry.            |
| Preconditions    | Module is available; a catalog response or safe cached list exists. |

| Field             | Record                                                       |
|-------------------|--------------------------------------------------------------|
| Success guarantee | A labeled tournament/team context is selected and preserved. |
| Priority          | High                                                         |
| Frequency         | Each new planning context                                    |

## Main flow

1. Open the tournament selector.
2. Review searchable tournament and team labels.
3. Choose a tournament and optional team.
4. Confirm the selected context.
5. Load the ticket-state comparison for that context.

## Alternate flows

- A1 at step 2: Catalog is unavailable; show the last timestamped list and a retry action.
- A2 at step 3: No context matches; clear filters and preserve the query.

## Exception flows

- E1: Network loss retains the selection locally and labels the state offline.
- E2: Login mismatch offers account switch without exposing another account's plan.

| Field          | Record                                                                                                                                                  |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Business rules | Tournament identity and freshness must be visible; no state is inferred from an absent catalog row.                                                     |
| Data           | Tournament ID, team ID, locale, selected context, timestamp.                                                                                            |
| Usability      | Recognition over recall; selected context stays visible.                                                                                                |
| Accessibility  | Keyboard-operable search and selection; labels are announced.                                                                                           |
| Privacy        | Selection may remain local until the user explicitly saves it.                                                                                          |
| Traceability   | PA1 FIFA drawback -> F2-E03/E04/E09 -> F-SES-03/04 -> UR-P-F2 -> UR-F03 -> Mobile overview -> 1 vote -> TP-FIFA -> F-A1 -> PA3 context-resumption test. |

FIFA

F-UC01 • Choose tournament context

**FIFA World Cup 2026**

United States, Canada & Mexico Selected  
context persists on return

SELECTED

**Women's World Cup 2027**

Brazil 24 June - 25 July 2027

**Context summary**

Tournament: FIFA World Cup 2026 • Team: Any • Locale: United States

Compare ticket states

Figure UC-VIS-F-UC01. F-UC01 mini prototype for its specific task, showing concrete data, the primary action, and an explicit recovery or boundary cue.



## 2.2 F-UC02 - Compare ticket states

| Field             | Record                                                                                                                  |
|-------------------|-------------------------------------------------------------------------------------------------------------------------|
| ID                | F-UC02                                                                                                                  |
| Goal              | Compare ticket states                                                                                                   |
| Scope             | FIFA Ticket Confidence Module                                                                                           |
| Primary actor     | Ticket planner                                                                                                          |
| Supporting actor  | Governed ticket-state feed                                                                                              |
| Trigger           | A tournament context is selected.                                                                                       |
| Preconditions     | Context is valid; state feed may return current, stale, missing, sold out, resale, waiting room, or hospitality states. |
| Success guarantee | Available states are compared without treating missing data as an outcome.                                              |
| Priority          | Critical                                                                                                                |
| Frequency         | Every planning visit                                                                                                    |

### Main flow

1. Request states for the selected context.
2. Display each channel with a text state label.
3. Compare primary sale, resale, hospitality, wait, and interest options.
4. Explain unavailable and sold-out distinctions.
5. Choose a state to inspect or stay on the comparison.

### Alternate flows

- A1 at step 2: State is stale; retain it with timestamp and warning.
- A2 at step 2: State is missing; show unknown and retry, not sold out.
- A3 at step 3: Waiting room is active; explain temporary queue behavior.

### Exception flows

- E1: Feed conflict shows both governed sources and blocks a confident action.
- E2: Network loss retains the last labeled snapshot.

| Field          | Record                                                                           |
|----------------|----------------------------------------------------------------------------------|
| Business rules | Sold out, resale, waiting room, missing, and stale are mutually explicit labels. |

| Field         | Record                                                                                                                                                                          |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data          | Context, channel, state, source, timestamp, eligibility note.                                                                                                                   |
| Usability     | Visibility of system status and error prevention.                                                                                                                               |
| Accessibility | State uses text and icon, never color alone; table headers are semantic.                                                                                                        |
| Privacy       | No personal data required for comparison.                                                                                                                                       |
| Traceability  | PA1 FIFA ticket drawback -> F2-E09 -> F-SES-04/05/06 -> UR-P-F1 -> UR-F01 -> Ticket status clarity -> 10 votes -> TP-FIFA -> F-A1 -> recommendation -> PA3 state-accuracy test. |

FIFA
F-UC02 • Compare ticket states

FIFA World Cup 2026 • Group Stage, Match 17

**Official**
COMING SOON

COMING SOON 20:30 State and channel remain distinct

**Resale**
NOT OPEN

NOT OPEN 20:30 State and channel remain distinct

**Hospitality**
AVAILABLE

AVAILABLE Partner route State and channel remain distinct

Inspect selected state

Figure UC-VIS-F-UC02. F-UC02 mini prototype for its specific task, showing concrete data, the primary action, and an explicit recovery or boundary cue.

## 2.3 F-UC03 - Inspect freshness

| Field             | Record                                                  |
|-------------------|---------------------------------------------------------|
| ID                | F-UC03                                                  |
| Goal              | Inspect freshness                                       |
| Scope             | FIFA Ticket Confidence Module                           |
| Primary actor     | Ticket planner                                          |
| Supporting actor  | Governed ticket-state feed                              |
| Trigger           | User opens freshness details for a ticket state.        |
| Preconditions     | A state record exists, even if stale or missing.        |
| Success guarantee | Timestamp, source, and confidence boundary are visible. |
| Priority          | Critical                                                |
| Frequency         | At important sale decisions                             |

### Main flow

1. Open freshness details.
2. Read last-updated time and source.
3. Compare current time with the freshness policy.
4. Review stale or missing explanation.
5. Refresh or return to comparison.

### Alternate flows

- A1 at step 3: Record is stale; block unqualified action and offer refresh.
- A2 at step 4: No update exists; show unknown and escalation guidance.

### Exception flows

- E1: Refresh fails; keep prior timestamp and recovery action.

| Field          | Record                                                                  |
|----------------|-------------------------------------------------------------------------|
| Business rules | A refresh attempt does not erase a valid prior timestamp.               |
| Data           | State ID, observed time, source ID, freshness threshold.                |
| Usability      | Clear feedback and calibrated confidence.                               |
| Accessibility  | Time is readable in local format and available to assistive technology. |

| Field        | Record                                                                                                                                        |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Privacy      | No additional personal data.                                                                                                                  |
| Traceability | PA1 evidence rules -> F2-E09 -> F-SES-04/06 -> UR-P-F1 -> UR-F01 -> Freshness and alerts -> 3 votes -> TP-FIFA -> F-A1 -> PA3 freshness test. |

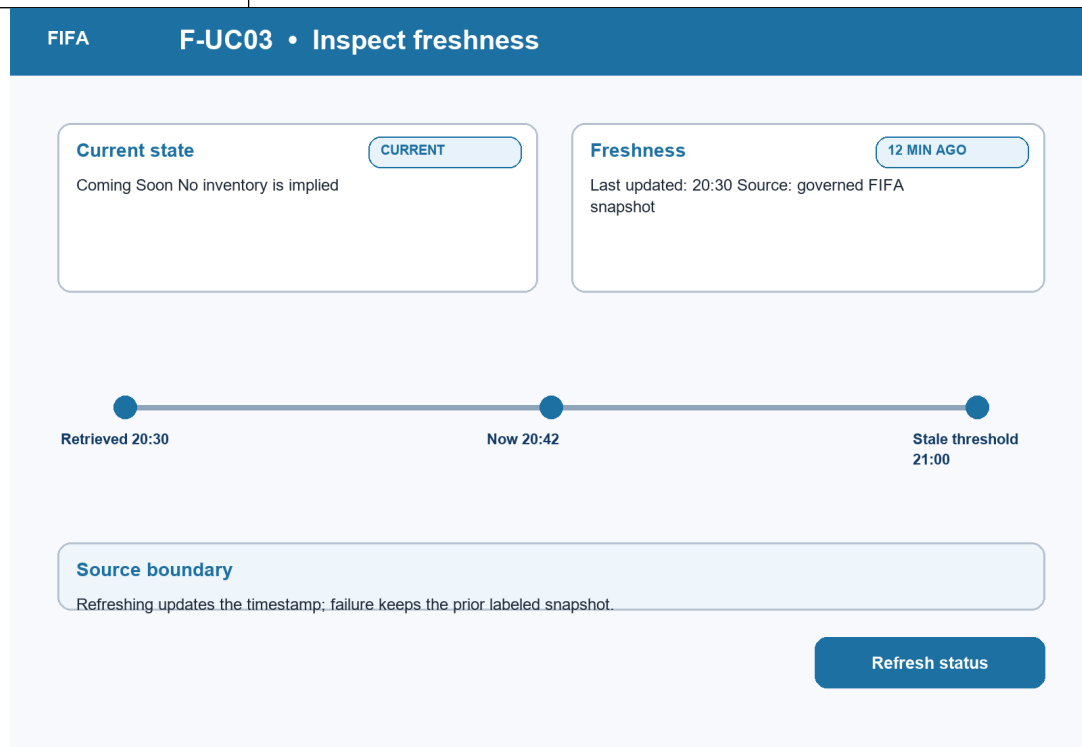


Figure UC-VIS-F-UC03. F-UC03 mini prototype for its specific task, showing concrete data, the primary action, and an explicit recovery or boundary cue.

## 2.4 F-UC04 - Preview destination

| Field             | Record                                                                                 |
|-------------------|----------------------------------------------------------------------------------------|
| ID                | F-UC04                                                                                 |
| Goal              | Preview destination                                                                    |
| Scope             | FIFA Ticket Confidence Module                                                          |
| Primary actor     | Ticket planner                                                                         |
| Supporting actor  | Official ticketing partner                                                             |
| Trigger           | User selects an outbound ticket action.                                                |
| Preconditions     | Destination metadata exists or the action remains blocked safely.                      |
| Success guarantee | Domain, purpose, context transfer, and return behavior are previewed before departure. |
| Priority          | Critical                                                                               |
| Frequency         | Before each outbound action                                                            |

### Main flow

1. Open destination preview.
2. Read official partner name, domain, and purpose.
3. Review what context will transfer.
4. Choose Continue, Stay, or Return.
5. If Continue is chosen, open the destination with a return route.

### Alternate flows

- A1 at step 2: Metadata is partial; offer Stay and report the limitation.
- A2 at step 4: User stays; preserve comparison context.

### Exception flows

- E1: Partner failure returns the user to the preserved state with an explanation.

| Field          | Record                                                     |
|----------------|------------------------------------------------------------|
| Business rules | The module never claims a partner transaction is complete. |
| Data           | Partner ID, domain, purpose, context token, return URI.    |
| Usability      | User control, trust calibration, and error prevention.     |

| Field         | Record                                                                                                                                                                   |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Accessibility | Destination and choices have explicit labels and logical focus order.                                                                                                    |
| Privacy       | Transfer only required context after consent; do not expose plan data in a URL.                                                                                          |
| Traceability  | PA1 handoff drawback -> F2-E09 -> F-SES-02/05 -> UR-P-F1 -> UR-F02 -> Handoff trust -> 8 votes -> TP-FIFA -> F-A1 -> recommendation -> PA3 destination-recognition test. |

FIFA
F-UC04 • Preview destination

[tickets.fifa.com](#)

EXTERNAL

Official ticketing destination Purpose: registration and purchase Account: FIFA ticketing account may be required

**Context transfer**

Tournament and match are passed only after Continue. Stay keeps the comparison. Return never proves a purchase.

Stay

Continue

Figure UC-VIS-F-UC04. F-UC04 mini prototype for its specific task, showing concrete data, the primary action, and an explicit recovery or boundary cue.

## 2.5 F-UC05 - Subscribe to alert

| Field             | Record                                                                         |
|-------------------|--------------------------------------------------------------------------------|
| ID                | F-UC05                                                                         |
| Goal              | Subscribe to alert                                                             |
| Scope             | FIFA Ticket Confidence Module                                                  |
| Primary actor     | Ticket planner                                                                 |
| Supporting actor  | Notification service                                                           |
| Trigger           | User chooses to follow a state change.                                         |
| Preconditions     | A context is selected; permission is not assumed.                              |
| Success guarantee | Alert preference is saved or a clear non-notification alternative is provided. |
| Priority          | High                                                                           |
| Frequency         | During wait/interest states                                                    |

### Main flow

1. Choose Subscribe to alert.
2. Select relevant state changes and channel.
3. Review frequency, privacy, and permission explanation.
4. Grant permission and confirm.
5. Edit, snooze, or stop the alert later.

### Alternate flows

- A1 at step 4: Notification is denied; keep a local saved plan and show manual refresh.
- A2 at step 2: No channel is available; offer in-module status check.

### Exception flows

- E1: Service failure preserves unsent preferences for retry.

| Field          | Record                                                                                   |
|----------------|------------------------------------------------------------------------------------------|
| Business rules | No alert is enabled without affirmative consent; every alert has an unsubscribe control. |
| Data           | Context, selected changes, channel, consent timestamp, retention state.                  |
| Usability      | Control, prevention of notification fatigue, and clear status.                           |

| Field         | Record                                                                                                                                                                         |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Accessibility | Permission explanation is plain text; controls are keyboard operable.                                                                                                          |
| Privacy       | Minimize contact data and disclose retention.                                                                                                                                  |
| Traceability  | PA1 ticket uncertainty -> F2-E09 -> F-SES-06 -> UR-P-F1 -> UR-F01 -> Freshness and alerts -> 3 votes -> TP-FIFA -> F-A3 -> retained PA3 prototype -> notification-denial test. |

FIFA
F-UC05 • Set a ticket alert

Tournament

FIFA World Cup 2026

Changes

On sale • Resale opens

Email

minh.anh@example.com

Privacy

Edit, snooze, or stop the alert at any time.

Permission
OPTIONAL

Browser alerts OFF  
Fallback: saved on-site reminder

Save alert

Figure UC-VIS-F-UC05. F-UC05 mini prototype for its specific task, showing concrete data, the primary action, and an explicit recovery or boundary cue.



## 2.6 F-UC06 - Continue, stay or return

| Field             | Record                                                               |
|-------------------|----------------------------------------------------------------------|
| ID                | F-UC06                                                               |
| Goal              | Continue, stay or return                                             |
| Scope             | FIFA Ticket Confidence Module                                        |
| Primary actor     | Ticket planner                                                       |
| Supporting actor  | Official ticketing partner; identity service                         |
| Trigger           | User acts from destination preview or reopens a return route.        |
| Preconditions     | Comparison context exists; any return token is validated.            |
| Success guarantee | User remains oriented and partner-side completion is never inferred. |
| Priority          | Critical                                                             |
| Frequency         | Each handoff and return                                              |

### Main flow

1. Choose Continue, Stay, or Return.
2. If Continue, transfer only consented context.
3. If Stay, preserve the comparison state.
4. If Return, validate token or identity.
5. Restore context, freshness, and available actions.

### Alternate flows

- A1 at step 4: Token expired; offer safe account lookup or plan recreation.
- A2 at step 5: Login mismatch; switch account without exposing stored data.

### Exception flows

- E1: Partner or network failure restores the last safe state.

| Field          | Record                                                                         |
|----------------|--------------------------------------------------------------------------------|
| Business rules | A return is not proof of purchase; tokens are time-limited and single-purpose. |
| Data           | Context, return token, account reference, last state and timestamp.            |

| Field         | Record                                                                                                                                                  |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Usability     | Recovery, memorability, and persistent orientation.                                                                                                     |
| Accessibility | Restoration is announced without unexpected focus movement.                                                                                             |
| Privacy       | Validate identity before revealing saved plans.                                                                                                         |
| Traceability  | PA1 handoff drawback -> F2-E09 -> F-SES-02/03/05/06 -> UR-P-F1/F2 -> UR-F02/F03 -> Recovery and return -> 1 vote -> TP-FIFA -> F-A1 -> PA3 return test. |

FIFA
F-UC06 • Continue, stay, or return

**Decision summary**
READY

FIFA World Cup 2026 • Match 17 • Party 2 Destination: tickets.fifa.com • Status checked 20:30

**CONTINUE**
Open the official destination with consented context.

**STAY**
Remain on FIFA.com with the comparison preserved.

**RETURN**
Validate identity and restore the last safe state.

A return is not confirmation of a purchase.

Choose action

Figure UC-VIS-F-UC06. F-UC06 mini prototype for its specific task, showing concrete data, the primary action, and an explicit recovery or boundary cue.

## 2.7 C-UC01 - Open game

| Field             | Record                                                                                |
|-------------------|---------------------------------------------------------------------------------------|
| ID                | C-UC01                                                                                |
| Goal              | Open game                                                                             |
| Scope             | Chess Guided Beginner Review Module                                                   |
| Primary actor     | Beginner / returning learner                                                          |
| Supporting actor  | Game store                                                                            |
| Trigger           | User chooses a recent game, PGN import, or board setup.                               |
| Preconditions     | A source route is available; input may be absent or invalid.                          |
| Success guarantee | A valid review artifact is opened or input is preserved with a clear correction path. |
| Priority          | High                                                                                  |
| Frequency         | Each review                                                                           |

### Main flow

1. Choose Recent game, Import PGN, or Set up position.
2. Select or enter the game artifact.
3. Confirm side to move and board orientation.
4. Validate notation and access rights.
5. Open the review entry.

### Alternate flows

- A1 at step 1: No game exists; emphasize import/setup routes.
- A2 at step 4: PGN is invalid; identify the first error and preserve input.

### Exception flows

- E1: Mobile interruption preserves the draft locally.

| Field          | Record                                                                           |
|----------------|----------------------------------------------------------------------------------|
| Business rules | Private games follow account permissions; invalid PGN is never silently changed. |
| Data           | Game ID, PGN, ownership, side to move, orientation.                              |
| Usability      | Clear setup choices and recoverable validation.                                  |
| Accessibility  | Text notation alternative and fully labeled board                                |

| Field        | Record                                                                                                                                                     |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | controls.                                                                                                                                                  |
| Privacy      | Respect game visibility and minimize copied metadata.                                                                                                      |
| Traceability | PA1 analysis drawback -> C2-E10 -> C-SES-01/02/05 -> UR-P-C1 -> UR-C01 -> Review setup -> 3 votes -> TP-CHESS -> C-A1 -> recommendation -> PA3 setup test. |

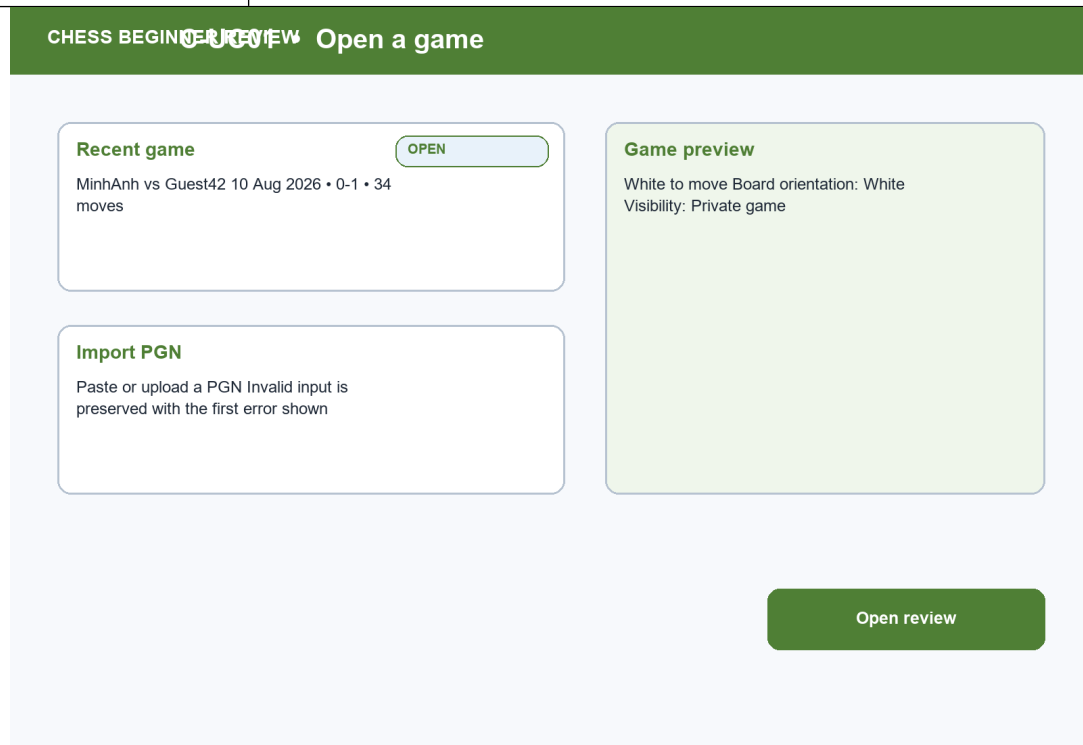


Figure UC-VIS-C-UC01. C-UC01 mini prototype for its specific task, showing concrete data, the primary action, and an explicit recovery or boundary cue.

## 2.8 C-UC02 - Select beginner review

| Field             | Record                                                                     |
|-------------------|----------------------------------------------------------------------------|
| ID                | C-UC02                                                                     |
| Goal              | Select beginner review                                                     |
| Scope             | Chess Guided Beginner Review Module                                        |
| Primary actor     | Beginner / returning learner                                               |
| Supporting actor  | Analysis service                                                           |
| Trigger           | A valid game artifact is opened.                                           |
| Preconditions     | Review entry is visible; premium/access restrictions may apply.            |
| Success guarantee | The beginner preset is selected without removing advanced analysis access. |
| Priority          | Critical                                                                   |
| Frequency         | Each beginner review                                                       |

### Main flow

1. Open review choices.
2. Read Beginner Review purpose and scope.
3. Select Beginner Review.
4. Confirm the game and perspective.
5. Start the guided review.

### Alternate flows

- A1 at step 2: Terminology help is requested; show inline explanation.
- A2 at step 3: Premium restriction applies; explain available scope and alternatives.

### Exception flows

- E1: Analysis service unavailable; offer retry and practice from the unreviewed game.

| Field          | Record                                                                                 |
|----------------|----------------------------------------------------------------------------------------|
| Business rules | Restrictions are stated before commitment; expert controls remain available by choice. |
| Data           | Game artifact, preset, perspective, entitlement.                                       |
| Usability      | Recognition over recall and progressive disclosure.                                    |

| Field         | Record                                                                                                                                                                     |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Accessibility | Keyboard-only selection and focus-visible controls.                                                                                                                        |
| Privacy       | No extra personal data beyond game access.                                                                                                                                 |
| Traceability  | PA1 analysis drawback -> C2-E10 -> C-SES-01/02/05 -> UR-P-C1 -> UR-C01 -> Entry choice overload -> 4 votes -> TP-CHESS -> C-A1 -> recommendation -> PA3 entry-choice test. |



Figure UC-VIS-C-UC02. C-UC02 mini prototype for its specific task, showing concrete data, the primary action, and an explicit recovery or boundary cue.

## 2.9 C-UC03 - Identify mistake

| Field             | Record                                                                               |
|-------------------|--------------------------------------------------------------------------------------|
| ID                | C-UC03                                                                               |
| Goal              | Identify mistake                                                                     |
| Scope             | Chess Guided Beginner Review Module                                                  |
| Primary actor     | Beginner / returning learner                                                         |
| Supporting actor  | Analysis service                                                                     |
| Trigger           | Guided review completes its first pass.                                              |
| Preconditions     | A review is available; there may be no major mistake or several comparable mistakes. |
| Success guarantee | One learning priority is shown with honest uncertainty.                              |
| Priority          | Critical                                                                             |
| Frequency         | Each review                                                                          |

### Main flow

1. Receive the guided review.
2. Show the most important mistake or state that no major mistake exists.
3. Display the position and move context.
4. Explain why this moment matters in plain language.
5. Offer Try a better move.

### Alternate flows

- A1 at step 2: No major mistake; show a useful improvement instead.
- A2 at step 2: Multiple mistakes are comparable; explain the selection and allow comparison.

### Exception flows

- E1: Engine result is partial; label uncertainty and avoid a definitive claim.

| Field          | Record                                                          |
|----------------|-----------------------------------------------------------------|
| Business rules | One priority does not imply all other moves are irrelevant.     |
| Data           | Position, move, evaluation change, confidence, explanation key. |
| Usability      | Effectiveness, clear hierarchy, and calibrated status.          |

| Field         | Record                                                                                                                                                                                 |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Accessibility | Text description accompanies board highlight; no color-only meaning.                                                                                                                   |
| Privacy       | Game visibility rules continue to apply.                                                                                                                                               |
| Traceability  | PA1 learning/analysis gap -> C2-E10 -> C-SES-02/04/06 -> UR-P-C1/C2 -> UR-C02 -> Learning priority -> 3 votes -> TP-CHESS -> C-A1 -> recommendation -> PA3 mistake-comprehension test. |

CHESS BEGINNER REVIEW
C-UC03 Identify the key mistake

**Move 18: Ne5?** MISTAKE

Key mistake The knight leaves the queen unprotected.

**Why this matters**

Black can win material immediately. This is the clearest learning priority in the game.

**Uncertainty**

Other moves are available; one priority does not make them irrelevant.

Try a better move

Figure UC-VIS-C-UC03. C-UC03 mini prototype for its specific task, showing concrete data, the primary action, and an explicit recovery or boundary cue.



## 2.10 C-UC04 - Try better move

| Field             | Record                                                                     |
|-------------------|----------------------------------------------------------------------------|
| ID                | C-UC04                                                                     |
| Goal              | Try better move                                                            |
| Scope             | Chess Guided Beginner Review Module                                        |
| Primary actor     | Beginner / returning learner                                               |
| Supporting actor  | Analysis service                                                           |
| Trigger           | User chooses Try a better move.                                            |
| Preconditions     | A review position and legal move set exist.                                |
| Success guarantee | The attempted move receives feedback without losing the original position. |
| Priority          | High                                                                       |
| Frequency         | At each selected mistake                                                   |

### Main flow

1. Open the review position in trial mode.
2. Enter a legal candidate move.
3. Compare it with the learning priority.
4. Receive concise feedback.
5. Retry, reveal the recommended move, or return.

### Alternate flows

- A1 at step 2: Move is illegal; explain the rule and keep focus on the board.
- A2 at step 3: Move is also strong; acknowledge it and compare ideas.

### Exception flows

- E1: Engine unavailable; record the trial locally and defer comparison.

| Field          | Record                                                      |
|----------------|-------------------------------------------------------------|
| Business rules | User action in trial mode never changes the stored game.    |
| Data           | Position, candidate move, legal moves, comparison result.   |
| Usability      | Immediate feedback, user control, and safe experimentation. |
| Accessibility  | Keyboard coordinate entry and announced feedback.           |

| Field        | Record                                                                                                                                                      |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Privacy      | Trial data stays with the selected review unless saved.                                                                                                     |
| Traceability | PA1 analysis opportunity -> C2-E10 -> C-SES-02/06<br>-> UR-P-C1/C2 -> UR-C02 -> Learning priority -> 3<br>votes -> TP-CHESS -> C-A1 -> PA3 move-trial test. |

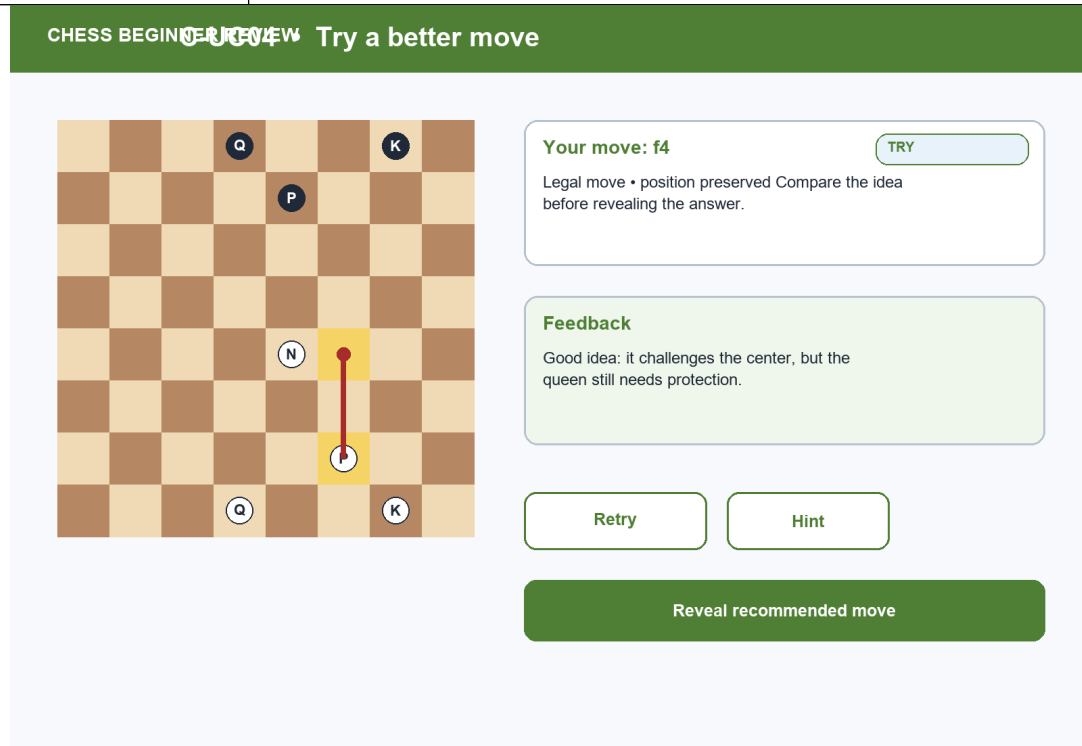


Figure UC-VIS-C-UC04. C-UC04 mini prototype for its specific task, showing concrete data, the primary action, and an explicit recovery or boundary cue.

## 2.11 C-UC05 - Read explanation

| Field             | Record                                                              |
|-------------------|---------------------------------------------------------------------|
| ID                | C-UC05                                                              |
| Goal              | Read explanation                                                    |
| Scope             | Chess Guided Beginner Review Module                                 |
| Primary actor     | Beginner / returning learner                                        |
| Supporting actor  | Explanation service                                                 |
| Trigger           | A mistake or better move is selected.                               |
| Preconditions     | An explanation exists or a safe fallback can describe the position. |
| Success guarantee | Plain language leads and terminology help is available on demand.   |
| Priority          | Critical                                                            |
| Frequency         | Each selected mistake                                               |

### Main flow

1. Open the explanation card.
2. Read the one-sentence learning point.
3. Review board evidence and move comparison.
4. Open terminology help if needed.
5. Choose Try again, Continue practice, or Advanced detail.

### Alternate flows

- A1 at step 2: Explanation is unavailable; show factual board changes and avoid invented rationale.
- A2 at step 4: Terminology help opens inline and returns focus to the term.

### Exception flows

- E1: Service failure keeps the move trial and offers retry.

| Field          | Record                                                               |
|----------------|----------------------------------------------------------------------|
| Business rules | Explanations distinguish fact, recommendation, and uncertainty.      |
| Data           | Learning point, board evidence, glossary terms, explanation version. |
| Usability      | Plain language, satisfaction, and recognition over                   |

| Field         | Record                                                                                                                                                                       |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | recall.                                                                                                                                                                      |
| Accessibility | Readable text, glossary semantics, and no forced motion.                                                                                                                     |
| Privacy       | No unrelated conversation or profile data.                                                                                                                                   |
| Traceability  | PA1 terminology drawback -> C2-E10 -> C-SES-01/02/04/06 -> UR-P-C1/C2 -> UR-C02 -> Vocabulary load -> 5 votes -> TP-CHESS -> C-A1 -> recommendation -> PA3 explanation test. |

CHESS BEGINNER REVIEW
Read the explanation

**Learning point**
PLAIN LANGUAGE

Protect the queen before starting a kingside attack.

**Mistake card**
MOVE 18

18.Ne5? The knight moved twice and stopped defending d3.

**Consequence**
WHY

...Qxd3 wins the queen. Term help: hanging piece = undefended piece.

Try again
Continue practice
Advanced detail

Figure UC-VIS-C-UC05. C-UC05 mini prototype for its specific task, showing concrete data, the primary action, and an explicit recovery or boundary cue.

## 2.12 C-UC06 - Continue to practice or advanced depth

| Field             | Record                                                                  |
|-------------------|-------------------------------------------------------------------------|
| ID                | C-UC06                                                                  |
| Goal              | Continue to practice or advanced depth                                  |
| Scope             | Chess Guided Beginner Review Module                                     |
| Primary actor     | Beginner / returning learner                                            |
| Supporting actor  | Learning catalog; analysis service                                      |
| Trigger           | User completes the explanation.                                         |
| Preconditions     | A learning point exists; practice or depth may be unavailable.          |
| Success guarantee | One relevant practice action is offered before optional advanced depth. |
| Priority          | Critical                                                                |
| Frequency         | Each completed explanation                                              |

### Main flow

1. Review the recommended lesson or puzzle.
2. Read why it matches the mistake.
3. Choose Continue to practice or Reveal advanced detail.
4. Open the chosen destination with review context.
5. Return to the review when desired.

### Alternate flows

- A1 at step 1: No exact practice exists; offer a broader topic and explain the gap.
- A2 at step 3: Advanced detail is restricted; explain scope without blocking practice.

### Exception flows

- E1: Mobile interruption preserves the review and selected next step.

| Field          | Record                                                                    |
|----------------|---------------------------------------------------------------------------|
| Business rules | Practice relevance is explainable; advanced depth is opt-in, not removed. |
| Data           | Learning point, catalog item, relevance reason,                           |

| Field         | Record                                                                                                                                                                                                 |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | disclosure state.                                                                                                                                                                                      |
| Usability     | Continuation, memorability, and progressive disclosure.                                                                                                                                                |
| Accessibility | Destination choice and return route are keyboard/screen-reader operable.                                                                                                                               |
| Privacy       | Share only the minimum review context with the learning catalog.                                                                                                                                       |
| Traceability  | PA1 learning strength + analysis drawback -> C2-E05/E07/E10 -> C-SES-03/04/06 -> UR-P-C2 -> UR-C03 -> Practice continuation -> 5 votes -> TP-CHESS -> C-A1 -> recommendation -> PA3 continuation test. |

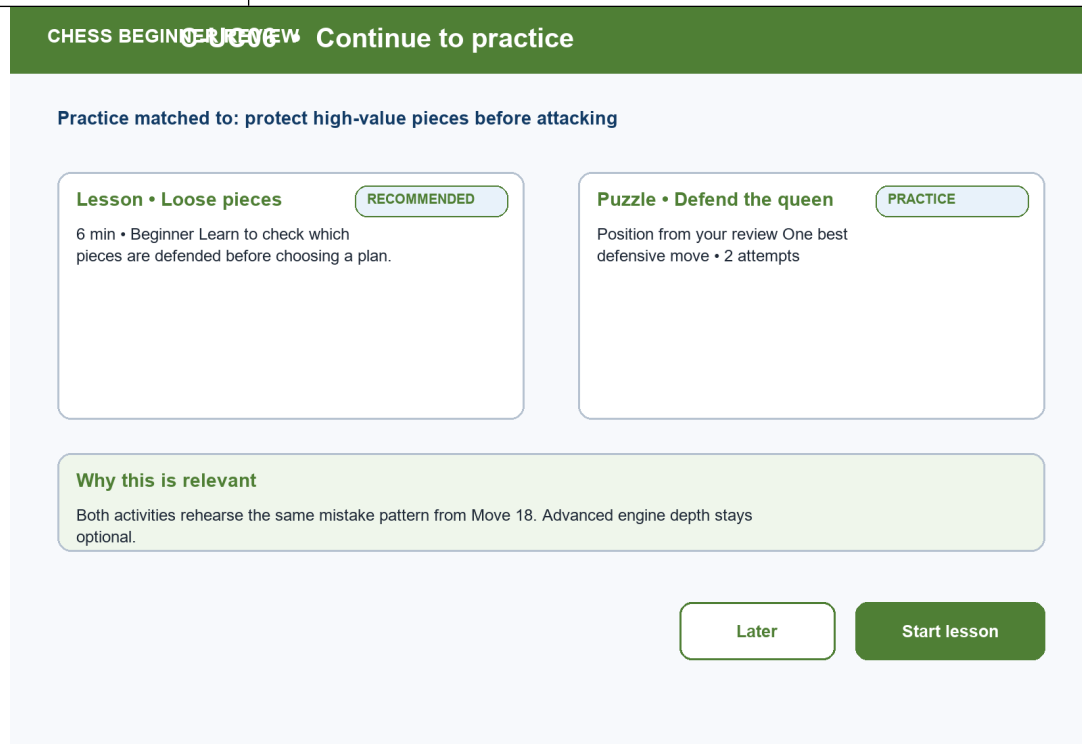


Figure UC-VIS-C-UC06. C-UC06 mini prototype for its specific task, showing concrete data, the primary action, and an explicit recovery or boundary cue.

### 3. Edge-case coverage

| Product | Required edge case                                         | Covered by        |
|---------|------------------------------------------------------------|-------------------|
| FIFA    | Stale state; missing state; sold out; resale; waiting room | F-UC02-F-UC03     |
| FIFA    | Partner failure; login mismatch;                           | F-UC01, F-UC04-F- |

| Product | Required edge case                                       | Covered by                   |
|---------|----------------------------------------------------------|------------------------------|
|         | notification denied; network loss                        | UC06                         |
| Chess   | No game; invalid PGN; engine unavailable                 | C-UC01-C-UC04                |
| Chess   | No major mistake; multiple mistakes; premium restriction | C-UC02-C-UC03                |
| Chess   | Mobile interruption; keyboard-only; terminology help     | C-UC01-C-UC02, C-UC05-C-UC06 |

## 4. Design rationale and validation target

**HCI rationale:** Course basis: LN01 - Introduction - v2.pdf, pp. 20, 22, 24; LN02 - Fundamental Concepts - Usability Dimensions\_2.pdf, pp. 62-67, 74-76, 83-84, 101-115; LN03 - UI Design Process.pdf, pp. 16-19, 23-24, 28-30, 34-36; LN04 - Task Analysis.pdf, pp. 3, 6-15, 39-49.

**PA3 validation target:** Test the six core steps per product and the stated alternate/exception flows without assuming implementation success.

## 5. State and Recovery Reference

### FIFA Ticket Concierge - Empty, Error, and Validation

| EMPTY                                                                                                                                                                                                                                                | ERROR                                                                                                                                                                                                                                                          | VALIDATION                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Trigger</b><br/>No tournament matches filters</p> <p><b>Message</b><br/>No matching tournament. Change tournament or date.</p> <p><b>User action</b><br/>Change filters</p> <p><b>Recovery</b><br/>Keep party size and eligibility answers</p> | <p><b>Trigger</b><br/>Status or partner service fails</p> <p><b>Message</b><br/>Ticket status is unavailable. No sale outcome is inferred.</p> <p><b>User action</b><br/>Retry or stay</p> <p><b>Recovery</b><br/>Show last update only when labeled stale</p> | <p><b>Trigger</b><br/>Tournament, email, or permission is missing</p> <p><b>Message</b><br/>Select a tournament and enter a valid email.</p> <p><b>User action</b><br/>Correct the field</p> <p><b>Recovery</b><br/>Offer on-site reminder when permission is denied</p> |

**Figure UC-STATE-FIFA.** FIFA empty, error, and validation reference

*Source: PA2 design QA artifact*

*Related: F-UC01-F-UC06*



## Chess Beginner Review - Empty, Error, and Validation

| EMPTY                                                                                                                                                                                                                                                        | ERROR                                                                                                                                                                                                                                                  | VALIDATION                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Trigger</b><br/>No recent game is available</p> <p><b>Message</b><br/>No game is ready for review. Import a PGN or play a game.</p> <p><b>User action</b><br/>Import game</p> <p><b>Recovery</b><br/>Return to recent activity after a game exists</p> | <p><b>Trigger</b><br/>Analysis service or network fails</p> <p><b>Message</b><br/>Analysis is unavailable. Your game remains saved.</p> <p><b>User action</b><br/>Retry analysis</p> <p><b>Recovery</b><br/>Open board-only review or return later</p> | <p><b>Trigger</b><br/>PGN or required input is missing</p> <p><b>Message</b><br/>Add a valid game or PGN before review starts.</p> <p><b>User action</b><br/>Fix input</p> <p><b>Recovery</b><br/>Keep valid fields and show the failing line</p> |

**Figure UC-STATE-CHESS.** Chess empty, error, and validation reference

Source: PA2 design QA artifact

Related: C-UC01-C-UC06

## 5.1 Complete Empty, Error, and Validation Matrix

| State                                    | Contract                                                                                                                                                                                                                                            |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FIFA - EMPTY - No tournament             | <p>Trigger: No tournament</p> <p>Message: No tournament matches the selected context.</p> <p>Action: Change tournament or date.</p> <p>Recovery: Keep party size and other valid answers.</p>                                                       |
| FIFA - EMPTY - No available ticket route | <p>Trigger: No available ticket route</p> <p>Message: No official route is available for this event.</p> <p>Action: Compare wait, interest, resale, or hospitality when allowed.</p> <p>Recovery: Return to the saved event without restarting.</p> |
| FIFA - ERROR - Ticket status unavailable | <p>Trigger: Ticket status unavailable</p> <p>Message: Ticket status is unavailable; no sale outcome is inferred.</p> <p>Action: Retry or stay.</p> <p>Recovery: Show a labeled last update only</p>                                                 |

| State                                       | Contract                                                                                                                                                                             |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | when available.                                                                                                                                                                      |
| FIFA - ERROR - Partner unavailable          | Trigger: Partner unavailable<br>Message: The official destination cannot respond.<br>Action: Stay on FIFA.com.<br>Recovery: Preserve context and retry later.                        |
| FIFA - ERROR - Network error                | Trigger: Network error<br>Message: The request failed before a result was confirmed.<br>Action: Retry the request.<br>Recovery: Resume at the state check with valid answers intact. |
| FIFA - VALIDATION - Tournament not selected | Trigger: Tournament not selected<br>Message: Select a tournament to continue.<br>Action: Choose a tournament.<br>Recovery: Return to the question step.                              |
| FIFA - VALIDATION - Invalid email           | Trigger: Invalid email<br>Message: Enter a valid email address.<br>Action: Correct the email.<br>Recovery: Retain all other answers.                                                 |
| FIFA - VALIDATION - Permission denied       | Trigger: Permission denied<br>Message: Browser alerts are off.<br>Action: Use an on-site reminder or change permission.<br>Recovery: Do not block the ticket decision.               |
| Chess - EMPTY - No game to review           | Trigger: No game to review<br>Message: No game is ready for review.<br>Action: Import a PGN or play a game.<br>Recovery: Return to review after a game exists.                       |
| Chess - EMPTY - No recent game              | Trigger: No recent game<br>Message: No recent game was found.<br>Action: Open the archive or import a game.<br>Recovery: Keep the chosen beginner-review mode.                       |
| Chess - ERROR - Analysis unavailable        | Trigger: Analysis unavailable<br>Message: Analysis is unavailable; the game remains saved.<br>Action: Retry analysis.<br>Recovery: Open board-only review or return later.           |

| State                                      | Contract                                                                                                                                                                |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chess - ERROR - Import failed              | Trigger: Import failed<br>Message: The game could not be imported.<br>Action: Review the PGN and retry.<br>Recovery: Keep the original input for correction.            |
| Chess - ERROR - Network error              | Trigger: Network error<br>Message: The analysis request failed.<br>Action: Retry when connected.<br>Recovery: Return to the saved game.                                 |
| Chess - VALIDATION - Invalid PGN           | Trigger: Invalid PGN<br>Message: The PGN contains an invalid or unsupported line.<br>Action: Correct the marked line.<br>Recovery: Retain valid metadata and moves.     |
| Chess - VALIDATION - Missing required game | Trigger: Missing required game<br>Message: Choose or import a game before review starts.<br>Action: Select a game.<br>Recovery: Return to the review entry.             |
| Chess - VALIDATION - Unsupported input     | Trigger: Unsupported input<br>Message: This input format is not supported.<br>Action: Use PGN or a recent saved game.<br>Recovery: Keep the current page and help link. |

## 6. Practical Difficulties and Lessons Learned

| Practical difficulty                                        | How the group handled it                                                                 | Lesson learned                                                 | Impact on the final version                                                         |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Use cases with similar verbs began to repeat the same path. | The group separated trigger, success guarantee, alternate flow, and visual for every ID. | A unique goal needs a unique state transition and visual.      | All twelve use cases now have a different visual beside the detailed specification. |
| Error branches were easy to hide in long tables.            | The group added a shared state reference while keeping case-specific exception text.     | Recovery must preserve valid context and name the next action. | FIFA and Chess now distinguish empty, service error, and validation failure.        |